Standard Operating Procedure
Installation of the RS-232C for
Y10-568Z00
<DBB-27, microcomputer278>

DOC. No. Z95-041900A

Install the option Y10-568Z00 to DBB-27 by following this procedure. After carry out procedure, please send back your check list to NIKKISO CO., LTD or local representative. **CAUTION)** Since the MICROCOMPUTER PCB and the corresponding COMMUNICATION PCB have been changed from the lot number "71104", please check the lot number before performing the work.

Manufacturer

NIKKISO CO.,LTD.

WARNING To reduce the risk of serious or fatal patient injury from device malfunction, installation and inspection must be performed only by trained technical staff.

1. Component parts

The followings are used to install Y10-568Z00 at DBB-27.

Table 1. Installation parts used for external output.

Item No.	Item name	QTY
X20-322W00	CABLE ASSY. (CABLE 26) <DCS-27> (Reference; Length 950mm)	1
X71-282Z00	PCB ASSY. (COMMUNICATION)	1
X71-111Z00	PCB ASSY. (CONNECTOR PLATE F)	1
B20-251A00	BRACKET (RS-232C) <DBB/DCS-27>	1
C02-225A00	CAP（RS-232C）<JST：J-DCES>	2
QC5MA2-4008ST	SEMS SCREW <M4*8L, with SW, W>	1
QC5MA2-3006ST	SEMS SCREW <M3*6L, with SW, W>	2
QSHM177	CIRCUIT BOARD SPACER	4

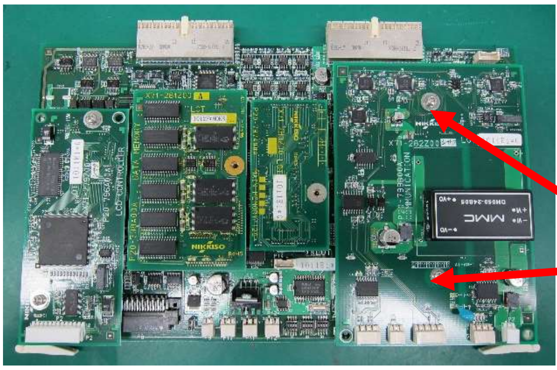
Table 2. Equipment used for operation check

Item No.	Item name			QTY
Y90-890Z00	Optional specification (BV download cable)			1
	Component parts	Z96-981A00	Standard operating procedure (BVM data download)	1
		X20-204W30	CABLE ASSY. (peripheral equipment) <L=3000MM>	1
—	PC ：Requirements: ① Hyper Terminal, communication application software for Microsoft Windows, is installed on the PC. ② RS-232C port is installed. (In the case of the PC with only a USB port, connect using a commercially USB to Serial conversion adapter (USB-RSAQ2: I-O DATA DEVICE, INC. *etc.*).)			1

2. Operational procedure

（1）Installation of PCB (COMMUNICATION)

①Press and hold the main power off switch of this device for 3 seconds or longer to turn off the power.
②Remove the left and right side covers.
③Remove the Upper back cover.
④Open the monitor cover and operation cover.
⑤Remove the cover (PCB) and pull out the PCB (MICROCOMPUTER) from the slot on the PCB (motherboard).
⑥Install the PCB (COMMUNICATION) to the PCB (MICOCOMPUTER).

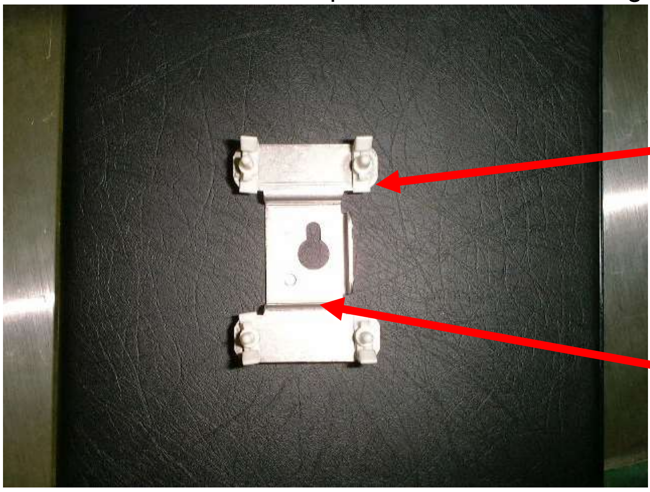


Connect the P1 connector of the PCB (COMMUNICATION) to the J13 connector of the PCB (MICROCOMPUTER).

Fix the PCB (COMMUNICATION) with screws (2 places).
M3×6L, 3 points sems

(2) Installation of printed circuit board (CONNECTOR PLATE F)

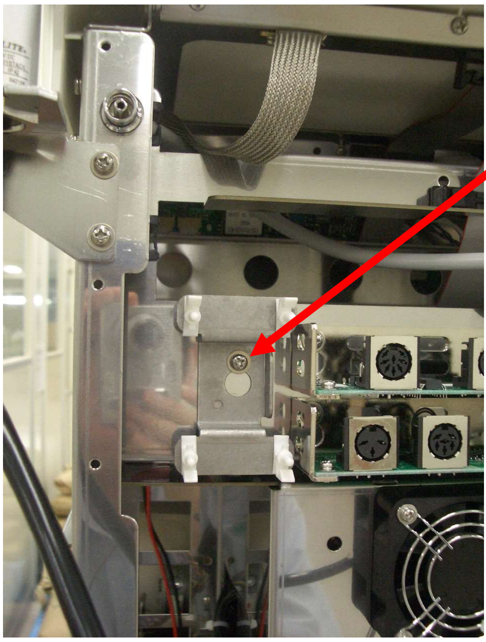
①Insert the circuit board spacer for board mounting into the bracket.



CIRCUIT BOARD SPACER
(4 places)

BRACKET

② Fix the bracket to the back of the device with screw.

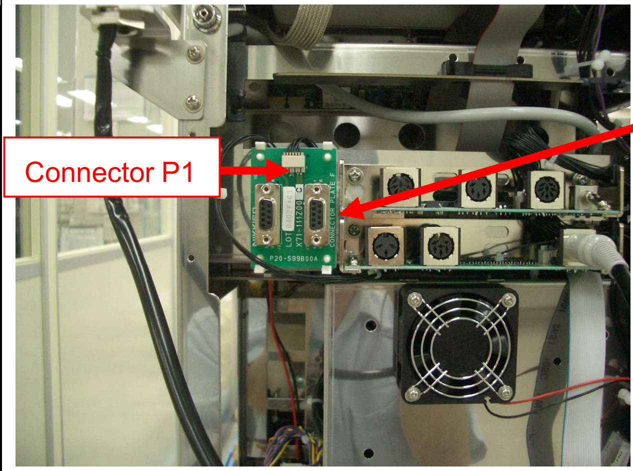


Fix the bracket with screw
M4×8L,3 points sems

【CAUTION】

1. After attaching the bracket, insert the PCB (CONNECTOR PLATE F).
2. When installing the bracket, do not pinch the lead wire.

③ Installation the PCB (CONNECTOR PLATE F) to the bracket.



Connector P1

Insert the PCB (CONNECTER PLATE F) into the circuit boat spacer.

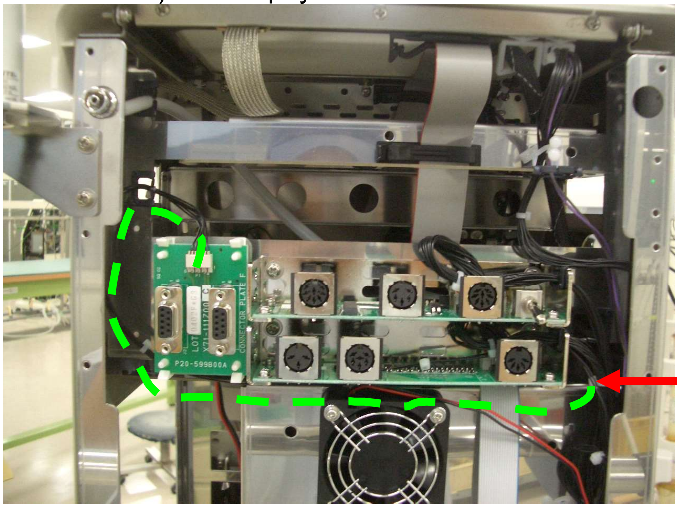
【CAUTION】

Pay attention to the position (direction) of connector P1

(3) Wiring of cable

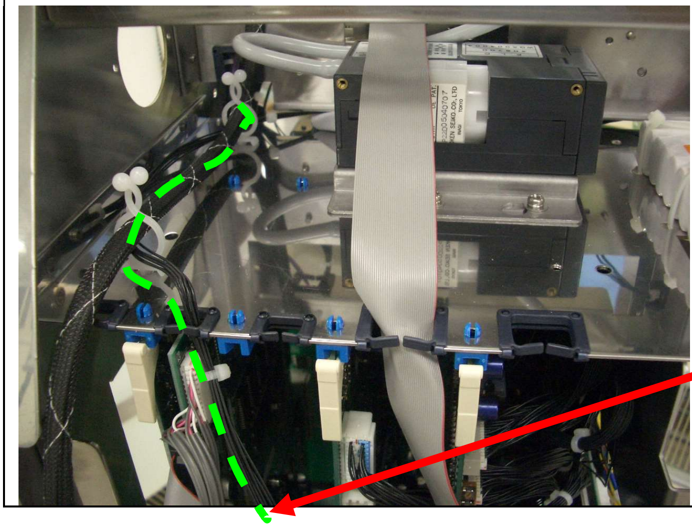
① Connect the connector of the CABLE ASSY. to connector P1 on the PCB (CONNECTER PLATE F).
CAUTION) Make sure the connector is firmly connected.

② Wire the CABLE ASSY. as shown below. (back side)
CAUTION) Please pay attention the cable shall not be pinched when wiring the cable.



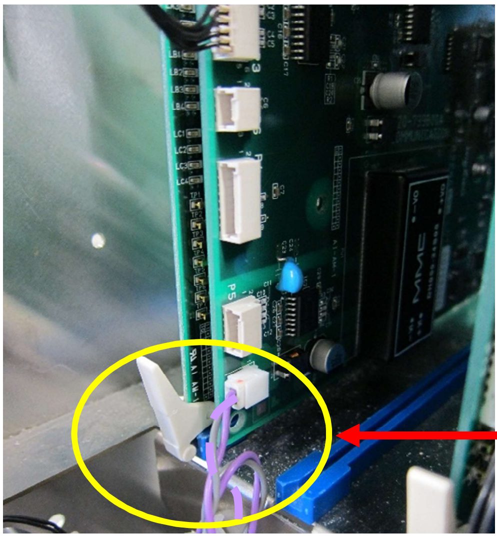
To the PCB (COMMUNICATION)

③ Wire the CABLE ASSY. As shown below. (left side)
CAUTION) Please pay attention the cable shall not be pinched when wiring the cable.



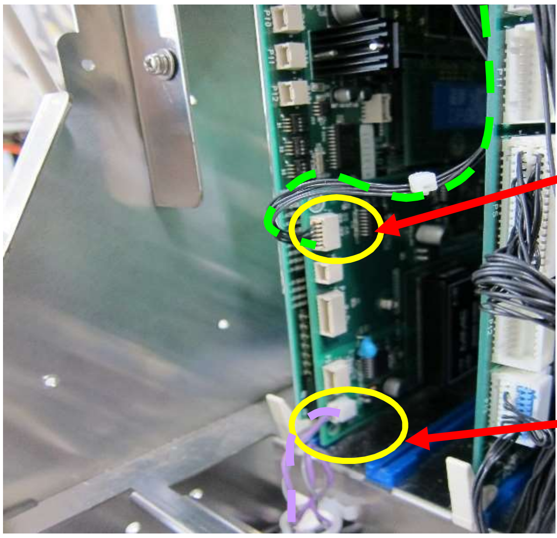
Pass the cable from the inside of the left side of the frame (electrical department) to the front.

④As shown below in the figure, extend the pre-wired cable (purple and gray) when wiring the one. (front)
CAUTION) Please pay attention the cable shall not be pinched when fixing it.



From the left side to the PCB (COMMUNICATION)

⑤Connect the cable to each connector of PCB (COMMUNICATION).
CAUTION) Make sure that the P2 and P3 connectors are firmly connected.
⑥Connect the PCB (MICROCOMPUTER) to the slot of the PCB (MOTHERBOARD).
CAUTION) Make sure the PCB (MICROCOMPUTER) is firmly connected to the slot on the PCB (MOTHERBOARD).



Connect the CABLE ASSY. to connector P3 on the PCB (COMMUNICATION).

Connect the cable (purple and gray) to the connector P2 on the PCB (COMMUNICATION).

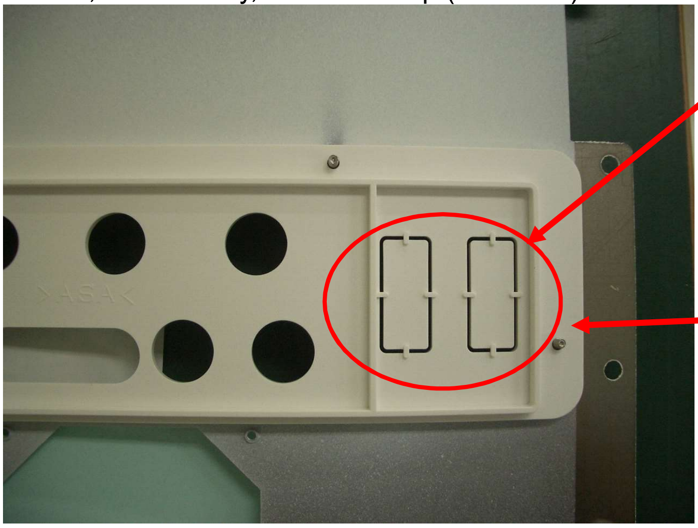
(4) Processing and installing of plate (back side connector)

①Open the spaces for the RS-232C connector on the plate (back side connector).

②After processing, install the upper back cover to this device.

CAUTION) Please pay attention the cable shall not be pinched.

And, if necessary, install the cap (RS-232C) <JST:J-DCES > to the RS-232C connector.



Cut 8 protrusions with nipper *etc.*.

Upper back cover plate (back connector)

(5) Microcomputer test

①Turn on the power to this device by pressing the main power ON switch.

②Remove the fuse (F3) of the power unit.

③Perform the "18.3 microcomputer test" in the adjusting mode and confirm that there are no abnormalities.

④Press and hold the main power off switch of this device for 3 seconds or longer to turn off the power.

(6) Operation check of external output 1 and external output 2

1) Preparation

①Connect the CABLE ASSY. (peripheral device) (DWG. No.; X20-204W30) to the "Output 1" connector of this device and the RS-232C port of the PC.

＊In the case of the PC with only a USB port, connect using a commercially USB to Serial conversion adapter.

②Turn on the power to this device by pressing the main power ON switch.

③Set "Model selection of separate BPM" of 37. blood pressure monitor from Setting 2 to "0: built-in".

＊ If the hospital setting value is not "0: built-in" (use an external blood pressure monitor), record the setting value.

＊To check the operation of external output 2, it is necessary to set "0: built-in".

④Turn on the computer and set up Hyper Terminal.

＊Start Device Manager, open Ports (COM and LPT), and check the COM number that can be used as a Communication port. However, when using the USB to Serial conversion adapter, connect the USB to Serial conversion adapter at first. (If you use a USB to Serial conversion adapter, it be displayed as Expansion port in Ports (COM and LPT).)

To start the device manager, open Settings (S) from Startup menu → Control Panel (C) → System , and click Device Manager (D) in Hardware tag.

＊According to the PC setting procedure of P2 from Standard operating procedure (Z96-981A00 BVM data download), perform the work. For the Connection method (N), select the COM number previously confirmed.

2) Operation check

①Enter "QWE" on the keyboard of your computer and check that "QWE" is displayed on the Hyper Terminal screen.

②Press and hold the main power off switch of this device for 3 seconds or longer to turn off the power.

③Remove the CABLE ASSY. (peripheral device) connected to the "output 1" connector of this device and connect it to the "output 2" connector of this device.

④Turn on the power to this device by pressing the main power ON switch.

⑤Enter "QWE" on the keyboard of your computer and check that "QWE" is displayed on the Hyper Terminal screen.

⑥Exit Hyper Terminal, turn off the power of the PC, and disconnect the CABLE ASSY. (peripheral device) from this device.

⑦If you change the "Model selection of separate BPM" of the 37.blood pressure monitor from Setting 2 to "0: built-in", enter the recorded setting value.

(7)**Installation of covers *etc.*.**

①Press and hold the main power off switch of this device for 3 seconds or longer to turn off the power.

②Attach the cover (PCB) and close the monitor cover and operation cover.

③Attach the left and right side covers.

CAUTION) Pay attention the cable shall not be pinched.

④If the external output 1 and external output 2 are not used for a long period of time, attach a cap (RS-232C) (DWG. No. ; C02-225A00) to the connector.

⑤Turn on the power to this device by pressing the main power ON switch.

CAUTION) Make sure there are no abnormalities.

3. Adjustment and Inspection after Installation

Inspection has to be performed after installation following below, and inspection check list is required to be submitted to NIKKISO CO., LTD or local representative.

Inspection check list

Refer to the Operating instruction and Technical Manual of DBB-27 to perform this inspection.

Yes☐: The inspection has been successfully finished or the measured value is within the criterion value.

No☐: The inspection has not been successfully finished or the measured value is not within the criterion value.

Model	DBB-27	Option	Y10-568Z00
Serial No.		Software version	Ver. .
Date		Name	

1.Check item before turning on power

		Y	N
PCB check ①	Make sure the PCB ASSY. (CONNECTOR PLATE F) is installed correctly.	☐	☐
PCB check ②	Make sure the PCB ASSY. (COMMUNICATION) is installed correctly.	☐	☐
Check the cable	Make sure the cable is installed correctly.	☐	☐
Check the plate	Make sure the plate (rear connector) is processed and installed correctly.	☐	☐
Installation of cover	Make sure the cover is attached correctly	☐	☐

2.Check item after turning on power

		Y	N
Microcomputer test	Perform a microcomputer test to confirm that there are no abnormalities.	☐	☐
Operation check of external output 1,2	Make sure the “QWE” and “RTY” is displayed correctly.	☐	☐

Location:____________ Org./Dept.____________________

Date:____________ Signature____________________

END